

ADVAIT PARULEKAR

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Education

PhD, UT Austin

Dept. of Electrical and Computer Engineering

January 2020-Present

(GPA 3.94/4.00)

BS, Texas A&M University

Dept. of Computer Science

August 2015-May 2019

(GPA: 4.00/4.00)

Experience

Research Intern, Apple

Data and Machine Learning Innovation

(Cupertino, USA. May 2025 - Sept 2025)

- Zero-Shot "Out of Distribution Detection" for on-device CLIP models.
- Improved accuracy at 20% FPR by 7% using density estimation techniques.

Research Intern, Google Research

Machine Learning and Optimization

(Bengaluru, India. June 2023 - Sept 2023)

- Studied simplicity bias in wide shallow neural networks.

Publications

- *Efficient Approximate Posterior Sampling with Annealed Langevin Monte Carlo.*
AP, Litu Rout, Karthikeyan Shanmugam, Sanjay Shakkottai.
to appear @ International Conference on Learning Representations (**ICLR 2026**).
- *In-Context Learning with Transformers: Softmax Attention Adapts to Function Lipschitzness.*
AP*, Liam Collins*, Aryan Mokhtari, Sujay Sanghavi, Sanjay Shakkottai. (co-first authors)
Spotlight @ Conference on Neural Information Processing Systems (**NeurIPS 2024**).
- *InfoNCE Loss Provably Learns Cluster-Preserving Representations.*
AP, Liam Collins, Karthikeyan Shanmugam, Aryan Mokhtari, Sanjay Shakkottai.
Conference on Learning Theory (**COLT 2023**).
- *PAC Generalization via Invariant Representations.*
AP, Karthikeyan Shanmugam, Sanjay Shakkottai.
Proceedings of the 40th International Conference on Machine Learning (**ICML 2023**).
- *Regret Bounds for Stochastic Shortest Path Problems with Linear Function Approximation.*
Daniel Vial, AP, Sanjay Shakkottai and R. Srikant.
Proceedings of the 39th International Conference on Machine Learning (**ICML 2022**).
- *Improved Algorithms for Misspecified Linear Markov Decision Processes.*
Daniel Vial, AP, Sanjay Shakkottai and R. Srikant.
Proceedings of the 25th International Conference on AI and Statistics (**AISTATS 2022**).
- *L1 Regression with Lewis Weights Subsampling.*
Aditya Parulekar, AP, Eric Price.
The International Conference on Randomization and Computation (**RANDOM 2021**).
- *A quadratically convergent iterative scheme for locating conical degeneracies in the spectra of parametric self-adjoint matrices.*
Gregory Berkolaiko, AP. (alphabetical order)
SIAM Journal on Matrix Analysis and Applications, 2021, Vol. 42, No. 1 : pp. 224-242.

Preprints

- *Diffusion-Based Posterior Sampling: A Feynman-Kac Analysis of Bias and Stability.*
Matias G. Delgadino, Sebastien Motsch, AP, William Porteous, Sanjay Shakkottai.
- *A Theoretical Justification for Image Inpainting using Denoising Diffusion Probabilistic Models.*
Litu Rout, AP, Constantine Caramanis, Sanjay Shakkottai.

- *Stochastic Linear Bandits with Protected Subspace.*
AP, Soumya Basu, Aditya Gopalan, Karthikeyan Shanmugam, Sanjay Shakkottai.

Competitions

2021 NSF Graduate Research Fellowship Program competition, Honorable Mention

2018 William Lowell Putnam Math Contest, Honorable Mention

2015 USA Physics Olympiad Team Member, 2013 Indian Math Olympiad Training Camp

Academic Experience

- **Service:** Reviewer @ AISTATS '21, 22; ICML '22, 23; RANDOM '22; NeurIPS '22, 24, 25; ICLR '26.
- **IPhO grader:** Invited to grade the 2022 International Physics Olympiad, Zurich, Switzerland.
- **Research Assistant:** Dept. of Mathematics, Texas A&M University - Aug 2019-Dec 2019.
- **SMaRT Camp Counsellor** - Dept. of Mathematics. Summers 2017, 2018, 2019.
- **Curriculum designer:** PhysicsWOOT @ Art of Problem Solving
- **Peer Teacher:** Departments of EE and CS @ TAMU. taught Data Structures and Algorithms, Circuits, Discrete Math, Probability, Operating Systems.

Graduate Coursework

Theory of Computation, Online Learning, Methods in Applied Mathematics (Functional Analysis), Stochastic Geometry, Stochastic Approximation, Advanced Probability, Sub-linear Algorithms, Combinatorial Optimization, Markov Chains and Mixing Times, Convex Optimization, Statistical Machine Learning, Game Theory, Information Theory, Cryptography, Randomized Algorithms, Algorithms in Structural Bioinformatics.